

```
# Exercise 1.1
# Create four variables about yourself

my_name = "Mucyo"
my_age = 21
my_gpa = 3.5
am_i_present = True

# Print each variable together with its type
print("Name:", my_name, "| Type:", type(my_name))
print("Age:", my_age, "| Type:", type(my_age))
print("GPA:", my_gpa, "| Type:", type(my_gpa))
print("Am I present:", am_i_present, "| Type:", type(am_i_present))
```

```
Name: Mucyo | Type: <class 'str'>
Age: 21 | Type: <class 'int'>
GPA: 3.5 | Type: <class 'float'>
Am I present: True | Type: <class 'bool'>
```

```
# Exercise 1.2

age_text = "21"

# Convert text to integer
age = int(age_text)

# Add 1
new_age = age + 1

# Print the result
print("Age after birthday:", new_age)
```

```
Age after birthday: 22
```

```
# Question 1.3

result = "3" + "4"

print(result)
```

```
34
```

```
# Exercise 2.1

my_name = "Mucyo"
my_age = 21

print(f"I am {my_name}, {my_age} years old, studying at AUCA.")
```

I am Mucyo, 21 years old, studying at AUCA.

Exercise 2.2

```
course = "Big Data Analytics"
```

```
# (a) Uppercase  
print("Uppercase:", course.upper())
```

```
# (b) Number of characters  
print("Number of characters:", len(course))
```

```
# (c) First 3 characters  
print("First 3 characters:", course[0:3])
```

Uppercase: BIG DATA ANALYTICS

Number of characters: 18

First 3 characters: Big

Exercise 3.1

```
scores = [72, 85, 91, 64, 78]
```

```
# (a) First and last score  
print("First score:", scores[0])  
print("Last score:", scores[-1])
```

```
# (b) Number of scores  
print("Number of scores:", len(scores))
```

```
# (c) Average  
average = sum(scores) / len(scores)  
print("Average:", average)
```

```
# (d) Highest and lowest score  
print("Highest score:", max(scores))  
print("Lowest score:", min(scores))
```

First score: 72

Last score: 78

Number of scores: 5

Average: 78.0

Highest score: 91

Lowest score: 64

Exercise 3.2

```
scores = [72, 85, 91, 64, 78]
```

```
# (a) Add 88 to the list
```

```
scores.append(88)

# (b) Replace position 3 with 66
scores[3] = 66

# (c) Print updated list
print("Updated scores:", scores)

# Calculate new average
new_average = sum(scores) / len(scores)

print("New average:", new_average)
```

```
Updated scores: [72, 85, 91, 66, 78, 88]
New average: 80.0
```

```
# Exercise 4.1

me = {
    "name": "Mucyo",
    "age": 21,
    "program": "Information Systems",
    "district": "Kigali"
}

print("Program:", me["program"])
```

```
Program: Information Systems
```

```
# Exercise 4.2

me = {
    "name": "Mucyo",
    "age": 21,
    "program": "Information Systems",
    "district": "Kigali"
}

# (a) Add score
me["score"] = 85

# (b) Increase age by 1
me["age"] = me["age"] + 1

# (c) Print all keys
print("Keys:", me.keys())

# Print the whole dictionary
print("Dictionary:", me)
```

```
Keys: dict_keys(['name', 'age', 'program', 'district', 'score'])
Dictionary: {'name': 'Mucyo', 'age': 22, 'program': 'Information Systems', 'dist
```

```
# Part 5 Dataset
# 20 AUCA students

students = [
    {
        "id": "AUCA001",
        "name": "Aline Uwase",
        "age": 21,
        "gender": "F",
        "program": "Information Systems",
        "district": "Gasabo",
        "attendance": 92,
        "score": 85
    },
    {
        "id": "AUCA002",
        "name": "Eric Niyonzima",
        "age": 23,
        "gender": "M",
        "program": "Software Engineering",
        "district": "Kicukiro",
        "attendance": 88,
        "score": 78
    },
    {
        "id": "AUCA003",
        "name": "Diane Mukamana",
        "age": 20,
        "gender": "F",
        "program": "Information Systems",
        "district": "Nyarugenge",
        "attendance": 95,
        "score": 91
    },
    {
        "id": "AUCA004",
        "name": "Jean Bosco Habimana",
        "age": 24,
        "gender": "M",
        "program": "Networking",
        "district": "Gasabo",
        "attendance": 70,
        "score": 64
    },
    {
        "id": "AUCA005",
        "name": "Clarisse Ingabire",
```

```
    "age": 22,
    "gender": "F",
    "program": "Accounting",
    "district": "Musanze",
    "attendance": 85,
    "score": 72
  },
  {
    "id": "AUCA006",
    "name": "Patrick Mugisha",
    "age": 21,
    "gender": "M",
    "program": "Software Engineering",
    "district": "Huye",
    "attendance": 90,
    "score": 83
  },
  {
    "id": "AUCA007",
    "name": "Sandrine Umutoni",
    "age": 19,
    "gender": "F",
    "program": "Information Systems",
    "district": "Kicukiro",
    "attendance": 97,
    "score": 94
  },
  {
    "id": "AUCA008",
    "name": "Emmanuel Nsengiyumva",
    "age": 25,
    "gender": "M",
    "program": "Networking",
    "district": "Rubavu",
    "attendance": 65,
    "score": 58
  },
  {
    "id": "AUCA009",
    "name": "Grace Uwimana",
    "age": 20,
    "gender": "F",
    "program": "Accounting",
    "district": "Gasabo",
    "attendance": 82,
    "score": 76
  },
  {
    "id": "AUCA010",
    "name": "Yves Ndayisenga",
    "age": 22,
```

```
    "gender": "M",
    "program": "Software Engineering",
    "district": "Nyarugenge",
    "attendance": 78,
    "score": 69
  },
  {
    "id": "AUCA011",
    "name": "Josiane Mukeshimana",
    "age": 21,
    "gender": "F",
    "program": "Information Systems",
    "district": "Muhanga",
    "attendance": 93,
    "score": 88
  },
  {
    "id": "AUCA012",
    "name": "Claude Iradukunda",
    "age": 23,
    "gender": "M",
    "program": "Networking",
    "district": "Gasabo",
    "attendance": 75,
    "score": 66
  },
  {
    "id": "AUCA013",
    "name": "Beata Nyirahabimana",
    "age": 24,
    "gender": "F",
    "program": "Accounting",
    "district": "Nyamagabe",
    "attendance": 89,
    "score": 81
  },
  {
    "id": "AUCA014",
    "name": "Olivier Twagirimana",
    "age": 20,
    "gender": "M",
    "program": "Software Engineering",
    "district": "Kicukiro",
    "attendance": 96,
    "score": 90
  },
  {
    "id": "AUCA015",
    "name": "Chantal Uwera",
    "age": 22,
    "gender": "F",
```

```
    "program": "Information Systems",
    "district": "Rusizi",
    "attendance": 68,
    "score": 61
  },
  {
    "id": "AUCA016",
    "name": "Fabrice Byiringiro",
    "age": 21,
    "gender": "M",
    "program": "Networking",
    "district": "Musanze",
    "attendance": 84,
    "score": 74
  },
  {
    "id": "AUCA017",
    "name": "Solange Ishimwe",
    "age": 19,
    "gender": "F",
    "program": "Software Engineering",
    "district": "Gasabo",
    "attendance": 91,
    "score": 87
  },
  {
    "id": "AUCA018",
    "name": "David Hakizimana",
    "age": 25,
    "gender": "M",
    "program": "Accounting",
    "district": "Nyagatare",
    "attendance": 72,
    "score": 55
  },
  {
    "id": "AUCA019",
    "name": "Immaculee Uwamahoro",
    "age": 23,
    "gender": "F",
    "program": "Information Systems",
    "district": "Kicukiro",
    "attendance": 87,
    "score": 79
  },
  {
    "id": "AUCA020",
    "name": "Samuel Rukundo",
    "age": 20,
    "gender": "M",
    "program": "Networking",
```


Lowest score: 55
Score range: 39

Exercise 5.2

```
# (a) First student
print("First student:")
print(students[0])

print()

# Last student
print("Last student:")
print(students[-1])

print()

# (b) Student at position 6
student = students[6]

print(
    f"{student['name']} from {student['district']} "
    f"studies {student['program']} and scored {student['score']}."
)
```

First student:
{'id': 'AUCA001', 'name': 'Aline Uwase', 'age': 21, 'gender': 'F', 'program': 'I

Last student:
{'id': 'AUCA020', 'name': 'Samuel Rukundo', 'age': 20, 'gender': 'M', 'program':

Sandrine Umutoni from Kicukiro studies Information Systems and scored 94.

```
districts.count("Gasabo")
districts.count("Kicukiro")
= districts.count("Nyarugenge")

from Gasabo: {gasabo_count}")
from Kicukiro: {kicukiro_count}")
from Nyarugenge: {nyarugenge_count}")
```

Students from Gasabo: 5
Students from Kicukiro: 4
Students from Nyarugenge: 3

Exercise 5.4

```
# Find the highest score
highest_score = max(all_scores)
```

```

highest_score = max(all_scores)

# Find its position
top_position = all_scores.index(highest_score)

# Find the student's name
top_name = names[top_position]

# Print the result
print(f"The top student is {top_name} with a score of {highest_score}%.")

```

The top student is Sandrine Umutoni with a score of 94.

```

# Exercise 5.4 Bonus

# TOP STUDENT

highest_score = max(all_scores)
top_position = all_scores.index(highest_score)
top_student = students[top_position]

print(
    f"The top student is {top_student['name']} with a score of {top_student['score']}% and attendance of {top_student['attendance']}%."
)

# LOWEST STUDENT

lowest_score = min(all_scores)
lowest_position = all_scores.index(lowest_score)
lowest_student = students[lowest_position]

print(
    f"The lowest student is {lowest_student['name']} with a score of {lowest_student['score']}% and attendance of {lowest_student['attendance']}%."
)

```

The top student is Sandrine Umutoni with a score of 94 and attendance of 97%.
The lowest student is David Hakizimana with a score of 55 and attendance of 72%.

